Joonyong Park

Ph.D. Student
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Education

Apr. 2025 ~ Present	**The University of Tokyo** *Ph.D. Student* Department of Information Physics & Computing, Graduate School of Information Science and Technology. **Research Theme:** Towards integrated audio generation model based on linguistic analysis of speech token and automatic evaluation of acoustic signals *Advisor: Yuki Saito*	Tokyo, Japan
Apr. 2023 ~ Mar. 2025	**The University of Tokyo** *Master's Degree* Department of Electrical Engineering & Information Systems, Graduate School of Engineering. **Thesis:** A Study on Analysis of Discrete Speech Representation Using Self-Supervised Learning Model and its Application to Speech Synthesis *Advisor: Daisuke Saito* *GPA: 3.0 / 3.0*	
Mar. 2016 ~ Feb. 2022	**Sogang University** *Bachelor's Degree* Major in Artificial Intelligence, Department of Electronic Engineering, School of Engineering. **Thesis:** Development of an Interactive Voice User Interface in Kiosk System Using Deep Learning *Advisor: Hyung-Min Park* *GPA: 3.54 / 4.3* (Military Service Period: Nov. 2017 ~ Aug. 2019)	Seoul, Korea

Research Interests

- Speech Synthesis / Speech Language Model
- Speech MOS Prediction System
- Audio Information Retrieval
- Self-Supervised Learning Models / Neural Audio Codecs
- Generative models (Autoregressive, Diffusion model)

Skills

Technologies

• Libraries	PyTorch, Transformers, librosa etc.
• Programming Languages	Python, C, MATLAB, shell etc.
• Software Development Tools	Git, Docker, Jupyter, Codex etc.

Spoken Languages

Korean (Native), English, Japanese (Near Native), Chinese (Fully Professional)

Academic Publications

Preprints

1. **Joonyong Park,** Shinnosuke Takamichi, David M. Chan, Shunsuke Kando, Yuki Saito, Hiroshi Saruwatari, *Language-Statistical Analysis of Neural Audio Codec Tokens Across Architectures, Corpora, and Noise Conditions,* Under review at IEEE TASLP.
2. **Joonyong Park**, Hounsu Kim, Yoonjeong Park, Ryota Kawamatsu, Wataru Nakata, Yuki Saito, Satoru Fukayama, Juhan Nam, *UTMOST: A Multi-Task Model for Assessing Emotional Speech Synthesis*, Under review at IEEE OJSP.
3. **Joonyong Park**, Jerry Li, *Character Identity is not Speaker Identity: KyaraBench and KyaraEmbed for Character Verification,* Under review at IEEE ICASSP 2027.
4. **Joonyong Park**, Jerry Li, *When the Persona Contradicts the Line: Diagnosing Plan-to-Synthesis Failures in Chain-of-Thought TTS*, Under review at IEEE ICASSP 2027.
5. Jungwoo Kim, **Joonyong Park**, Junyoung Koh, Jong-Seok Lee, *Neural Audio Codec for Robust Audio Deepfake Detection*, Under review at IEEE ICASSP 2027.
6. Hounsu Kim, **Joonyong Park**, Yuki Saito, Satoru Fukayama, Juhan Nam, *Beyond Token Revision: Investigating Mask-and-Replace Diffusion for Zero-Shot Text-to-Speech*, Under review at IEEE ICASSP 2027.
7. Hayeon Bang, Hounsu Kim, **Joonyong Park,** Jung-Jae Yu, Juhan Nam, *PitchIn: Natural-Language Vocal Coaching from Unstructured Community Feedback*, Under review at IEEE ICASSP 2027.
8. **Joonyong Park**, Jerry Li, *When Does Predictor-Based RL Align with Human Perception? A Study of Subjective Rewards in Codec-Based Speech Language Models*, Submitted to ACL ARR, 2026.
9. **Joonyong Park,** David M. Chan, Yuki Saito, Hiroshi Saruwatari, *Auditing Protocol-Level Shortcuts in Large Audio Language Model Judges for Speech Evaluation*, Submitted to IEEE SLT 2026.
10. David M. Chan, Rodolfo Corona, **Joonyong Park,** Cheol Jun Cho, Yutong Bai, Trevor Darrell, *Analyzing the Language of Visual Tokens*, Submitted to ACL AAR, 2025.

International Conferences (Peer-Reviewed)

1. **Joonyong Park**, Jerry Li, *AnimeScore: A Preference-Based Dataset and Framework for Evaluating Anime-Like Speech Style*, Proc. ISCA INTERSPEECH, 2026.

2. **Joonyong Park**, Jungwoo Kim, Junyoung Koh, Yuki Saito, *Probing Token Spaces under Generator Shift in AI-Generated Music Detection*, ICML 2026 Workshop on Machine Learning for Audio, 2026.
3. **Joonyong Park**, Daisuke Saito, Nobuaki Minematsu, *MixedG2P-T5: G2P-free Speech Synthesis for Mixed-script texts using Speech Self-Supervised Learning and Language Model*, Proc. APSIPA ASC, 2025.
4. **Joonyong Park,** Shinnosuke Takamichi, David M. Chan, Shunsuke Kando, Yuki Saito, Hiroshi Saruwatari, *Analysing the Language of Neural Audio Codecs*, Proc. IEEE ASRU, 2025.
5. **Joonyong Park,** Kenichi Nakamura, *EmoSSLSphere: Multilingual Emotional Speech Synthesis with Spherical Vectors and Discrete Speech Tokens*, Proc. ISCA SSW, 2025.
6. **Joonyong Park**, Daisuke Saito, Nobuaki Minematsu, *Analytic Study of Text-Free Speech Synthesis for Raw Audio using a Self-Supervised Learning Model*, Proc. APSIPA ASC, 2024.
7. Kentaro Onda, **Joonyong Park**, Nobuaki Minematsu, Daisuke Saito, *A Pilot Study of GSLM-based Simulation of Foreign Accentuation Only Using Native Speech Corpora*, Proc. ISCA INTERSPEECH, 2024.
8. Shinnosuke Takamichi, Hiroki Maeda, **Joonyong Park,** Daisuke Saito, Hiroshi Saruwatari, *Do learned speech symbols follow Zipf's law?,* Proc. IEEE ICASSP, 2024.
9. **Joonyong Park,** Shinnosuke Takamichi, Tomohiko Nakamura, Kentaro Seki, Detai Xin, Hiroshi Saruwatari, *How Generative Spoken Language Model Encodes Noisy Speech: Investigation from Phonetics to Syntactics*, Proc. ISCA INTERSPEECH, 2023.

Domestic Conferences (1st Author only)

1. **朴浚鎔**、Li Jerry、“ドメイン特化型の知覚的報酬を用いた強化学習によるアニメスタイル音声合成の最適化”、2026年音学シンポジアム発表論文集、2026.
2. **朴浚鎔**、高道慎之介、David M. Chan、神藤駿介、齋藤佑樹、猿渡洋、“ニューラルオーディオコーデックにおける雑音頑健性分析 ～Zipf則・Heaps則に基づく言語統計構造と劣化音声の関係～”、研究報告音声言語情報処理、2026.
3. **朴浚鎔**、Li Jerry、“AnimeMOS：「アニメ声らしさ」の客観評価に向けたデータとMOS予測モデルの設計”、日本音響学会2026年春季研究発表会講演論文集、2026.
4. **朴浚鎔**、陳晋升、土井啓成、朴炳宣、白旗悠真、橘健太郎、棟楊、齋藤佑樹、猿渡洋、“ ニューラルオーディオコーデック特徴量を用いた音声から話者特有の表情予測モデルの構築及び分析、日本音響学会2026年春季研究発表会講演論文集、2026.
5. **朴浚鎔**、高道慎之介、David M. Chan、神藤駿介、齋藤佑樹、猿渡洋、“音声トークンの言語に関する分析”、2025年音学シンポジアム発表論文集、2025.
6. **朴浚鎔**、中村建一、“言語別球面座標感情ベクトルと音声トークンを使用した多言語感情音声合成に関する考察”、日本音響学会2025年春季研究発表会講演論文集、2025.
7. **朴浚鎔**、齋藤大輔、峯松信明、“自己教師あり学習モデルと T5 モデルを用 い

たG2Pフリー音声合成に関する研究” 研究報告音声言語情報処理、2024.

8. **朴浚鎔**、齋藤大輔、峯松信明、“テキストのない音声に対する自己教師あり 学習モデルによる音声合成の分析 ～多言語活用を中心に～”、研究報告音声言語情報処理、2024.
9. **朴浚鎔**、齋藤大輔、峯松信明、“音声合成システムの入力表現に関する分析的検討”、日本音響学会2023年秋季研究発表会講演論文集、2023.
10. **朴浚鎔**、高道慎之介、中村友彦、関健太郎、辛徳泰、猿渡洋、“Generative Spoken Language Modelを用いた劣化雑音音声の分析と他言語への適用”、日本音響学会2023年春季研究発表会講演論文集、2023.
11. **Joonyong Park,** Jaehyeon Ko, Seok Hyun Kim, Won Jeong, Jiho Lee, Bon Hyeok Ku, Hyung-Min Park, *Development of an Interactive Voice User Interface in Kiosk System Using Deep Learning*, Proc. Conference of the Brain Engineering Society of Korea, 2022.
12. **박준용,** 이대호, 방윤, 이나연, 김지환, “KorQuAD 데이터셋과 KoBERT Tokenizer를 이용한 KoBART 모델 성능의 분석”, 한국컴퓨터종합학술대회 논문집, 2021.

Honors and Scholarships

Apr. 2026 ~ Mar. 2028	**Japan Society for the Promotion of Science Doctoral Course Research Fellowships (JSPS DC2)** 225,000 JPY/month + Research Funds	
Jun. 2025	**Best Paper Award** Sound Symposiums 2025	
Dec. 2024	**Best Student Paper Award (LY Corporation Award)** 154th Conference on Spoken Language Processing (SLP)	
Nov. 2022	**Honorable Mention Award** ETRI OpenAPI Competition	
Apr. 2022 ~ Mar. 2026	**Japanese Government (MEXT) Scholarship** 147,000 JPY/month	
May. 2021	**Best Undergraduate Paper Award** Korean Information Science Society	

Research Experiences

Jul. 2026 ~ Sep. 2026	**Music and Audio Computing Lab, KAIST** *Visiting Research Student* Research on vocal coaching models, fake music detection, and speech neural audio codecs.	Daejeon, Korea
Apr. 2025 ~ Mar. 2026	**LY Corporation** *Industry-Academic Collaborative Research* Research on generating speech-to-face expression generation	Tokyo, Japan

for virtual avatar.

Jun. 2024 ~ Jul. 2024	**Darrell Group, University of California, Berkeley** *Visiting Research Student* Research on analyzing the language of text-derived tokens from various modalities, including visual, auditory, and multimodal media.	California, USA
Apr. 2022 ~ Mar. 2023	**Saruwatari & Koyama Lab, The University of Tokyo** *Research Student* Research on self-supervised learning and generative model on speech synthesis for post-filter signal processing used in speech enhancement.	Tokyo, Japan
Oct. 2021 ~ Mar. 2022	**Intelligent Information Processing Lab, Sogang University** *Internship* Development of dialogue system and speech synthesis in *IIPERIA*: Intelligent Information Processing with End-2-end Representation for Interactive Assistant.	Seoul, Korea
Jan. 2021 ~ Jun. 2021	**Auditory Intelligent Lab, Sogang University** *Internship* Development of *SGRecognizer* Demo: Korean unlimited vocabulary speech recognition system using Kaldi.	Seoul, Korea

Work Experiences

Oct. 2025 ~ Present	**Spellbrush** *Researcher (Part-time)* Research on speech generation in multimodal dialogue models and video generation models.	Tokyo, Japan
Oct. 2024 ~ Jun. 2025	**Cygames Inc.** *Researcher (Part-time)* Research on real-time foley sound synthesis in the virtual environment.	Tokyo, Japan
Aug.2024 ~ Dec. 2024	**Preferred Networks Inc.** *Researcher (Internship)* Research on multilingual emotional speech synthesis with spherical vectors and discrete speech tokens.	Tokyo, Japan
Jan.2024 ~ Feb.2024	**NTT R&D** *Researcher (Internship)* Analysis towards G2P-free speech synthesis using text-speech token encoder through hidden unit T5.	Kanagawa, Japan
Aug.2023 ~ Oct. 2023	**DeNA Co., Ltd.** *AI Specialist (Internship)* Improvement and implementation of cross-lingual voice conversion for products.	Tokyo, Japan

Apr. 2023 ~ Mar. 2026	**CoeFont Co., Ltd.** *Researcher (Part-time)* Improvement and acceleration of the self-supervised learning framework to create natural and real-time speech synthesis and voice conversion.	Tokyo, Japan
Sep. 2020 ~ Jun. 2021	**Samsung Medical Center** *Research Assistant (Part-time)* Optimization of classification model and distillation of datasets for detecting diseases from cardiac MRI.	Seoul, Korea

Teaching and Educational Experience

Oct. 2023~ Jan. 2024	**Software II** *Teaching Assistant* Course for Second year students for Department of Electronic and Information Engineering, Faculty of Engineering, The University of Tokyo.	Tokyo, Japan

Other Projects & Experiences

Jan. 2025~ Sep. 2025	**jarre.ai** *Team Leader, Product Designer, AI Engineer* Student-based startup project, with theme of 'Text-based AI implementation solution tool for video games'. Selected project for 2025 UTokyo Spring Founders Program.	Tokyo, Japan